

ip Command Cheat Sheet

View and change Linux network interfaces, addresses and routes.

Using it

```
ip [options] OBJECT [command] ← command defaults to "show"
```

Object	Short	Shows	Option	What it does
link	ip l	Interfaces (ports, bridges, Wi-Fi)	-br	Brief, one line per interface
addr	ip a	IP addresses on each interface	-d	Details (type: bridge, vlan, bond...)
route	ip r	Routing table (where traffic goes)	-4 / -6	IPv4 only / IPv6 only
neigh	ip n	Known neighbors (IP → MAC)	-s	Traffic statistics (bytes, errors)
			-c	Color output

Everyday commands

Command	Use it to
ip -br addr	List every interface and its addresses (best first look)
ip addr show eno1	Full details for one interface
ip route get 1.1.1.1	Find the main interface and IP (the one used to reach the internet)
ip route show default	Show the default gateway (lowest metric wins if several)
ip link show type bridge	List all bridges
ip link show master main	List the interfaces plugged into bridge main
ip -d link show eno1	Show what eno1 belongs to (master main , bridge_slave)

Reading ip route get

```
1.1.1.1 via 192.168.1.1 dev main src 192.168.1.50
```

via = gateway (router) · **dev** = **main interface** · **src** = **main IP** · nothing is actually sent to 1.1.1.1

Reading ip addr

```
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 master main state UP
    link/ether 3c:ec:ef:12:34:56 brd ff:ff:ff:ff:ff:ff ← MAC address
    inet 192.168.1.50/24 brd 192.168.1.255 scope global dynamic ← IPv4 address / subnet size
    inet6 fe80::3eec:efff:fe12:3456/64 scope link ← IPv6 link-local (local network only)
```

You see	It means
UP	The interface is switched on
LOWER_UP	A cable or link is actually connected (missing = unplugged)
state DOWN / NO-CARRIER	Switched off / switched on but no cable or link
master main	This interface is a port of bridge (or bond) main ; its IP lives on main
dynamic	The address came from DHCP (no dynamic = set by hand)
lo, docker0, veth*, virbr0	Loopback or virtual interfaces; usually not the main one

Making changes (needs sudo)

Command	Does
ip addr add 10.0.0.5/24 dev eno1	Adds an IP address
ip addr del 10.0.0.5/24 dev eno1	Removes an IP address
ip link set eno1 up / down	Switches the interface on / off
ip route add default via 10.0.0.1	Sets the default gateway

Remember: changes made with **ip** are **lost on reboot**. To make them permanent, edit Netplan, NetworkManager, or **/etc/network/interfaces**.